

### 11.9. Functions Defined by Power Series.

Example: The Bessel function of order 0 is defined by.

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} (n!)^2}.$$

a). Find the domain of  $J_0$ .

b). Find the derivative of  $J_0$ .

Sol. Let  $a_n = \frac{(-1)^n x^{2n}}{2^{2n} (n!)^2}$

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{(-1)^{n+1} x^{2n+2}}{2^{2n+2} ((n+1)!)^2} \cdot \frac{2^{2n} (n!)^2}{(-1)^n x^{2n}} \right| \\ &= \frac{x^2}{4} \frac{(n!)^2}{((n+1)!)^2} \\ &= \frac{x^2}{4} \left( \frac{1}{n+1} \right)^2 \longrightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

So,  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1$  then  $\text{Dom } J_0 = \mathbb{R}$ .

b). 
$$\begin{aligned} \frac{d}{dx} (J_0(x)) &= \frac{d}{dx} \left( \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} (n!)^2} \right) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n 2n x^{2n-1}}{2^{2n} (n!)^2} \end{aligned}$$

### 11.10 Taylor and Maclaurin Series.

Def: Let consider a function such that.

$$f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n. \quad (M)$$

this is a Taylor series of  $f(x)$  centered at  $x=a$ .

if  $a=0$  then  $(M) = \sum_{n=0}^{\infty} C_n x^n$  is called a Maclaurin Series.

Thm. If  $f$  has a power series representation (expansion) at  $a$ , i.e.,

$$f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n \quad |x-a| < R.$$

then its coefficients are given by the formula

$$C_n = \frac{f^{(n)}(a)}{n!}$$

Rmk. 
$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f'''(a)}{3!} (x-a)^3 + \dots$$

Example: We know that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n.$$

$$\text{let } f(x) = \frac{1}{1-x} \quad ; \quad f(0) = 1.$$

$$f'(x) = \frac{1}{(1-x)^2} \quad ; \quad f'(0) = 1.$$

$$f''(x) = \frac{1 \cdot 2}{(1-x)^3} \quad ; \quad f''(0) = 1 \cdot 2$$

$$f'''(x) = \frac{1 \cdot 2 \cdot 3}{(1-x)^4} \quad ; \quad f'''(0) = 1 \cdot 2 \cdot 3.$$

$\vdots$

$$f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}} \quad ; \quad f^{(n)}(0) = n!$$

$$C_n = \frac{f^{(n)}(0)}{n!} = \frac{n!}{n!} = 1.$$

Example: For the function  $f(x) = e^x$ , find the Maclaurin series and its radius of convergence.

Sln.  $f(x) = e^x \quad f(0) = 1.$

$$f'(x) = e^x \quad f'(0) = 1.$$

$$f''(x) = e^x \quad f''(0) = 1.$$

$$f^{(n)}(x) = e^x \quad f^{(n)}(0) = 1.$$

then

$$e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n.$$

To find the radius of convergence,  $a_n = \frac{x^n}{n!}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \frac{|x|}{n+1} \longrightarrow 0 < 1.$$

then the radius is  $R = \infty$ .

Example. Find the Taylor Series for  $f(x) = e^x$  at  $a=2$ .

Sln:  $f(x) = e^x \Rightarrow f(2) = e^2$

$$f'(x) = e^x \Rightarrow f'(2) = e^2.$$

$$f^{(n)}(x) = e^x \Rightarrow f^{(n)}(2) = e^2.$$

then,

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!} (x-2)^n = \sum_{n=0}^{\infty} \frac{e^2}{n!} (x-2)^n.$$

$$a_n = \frac{e^2}{n!} (x-2)^n$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{e^2 (x-2)^{n+1}}{(n+1)!} \cdot \frac{n!}{e^2 (x-2)^n} \right| = \frac{|x-2|}{n+1} \xrightarrow[n \rightarrow \infty]{\text{as}} 0$$

then  $R = \infty$ .

Taylor Series of important functions.

1). Find the Maclaurin Series for  $\sin x$  and prove that it represents  $\sin x$  for all  $x$ .

Sln:

$$\begin{aligned}
 f(x) &= \sin x & f(0) &= 0. \\
 f'(x) &= \cos x & f'(0) &= 1 & f^{(2n+1)} \\
 f''(x) &= -\sin x & f''(0) &= 0. \\
 f'''(x) &= -\cos x & f'''(0) &= -1. \\
 f^{(4)}(x) &= \sin x & f^{(4)}(0) &= 0.
 \end{aligned}$$

Since the derivatives repeat in a cycle of four, we can write:

$$\begin{aligned}
 & f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots \\
 &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \quad \begin{cases} \text{if } n=2k \Rightarrow f^{(2k)}(0)=0 \\ \text{if } n=2k+1 \Rightarrow f^{(2k+1)}(0) \neq 0. \end{cases} \\
 &= \sum_{k=0}^{\infty} \frac{f^{(2k+1)}(0)}{(2k+1)!} x^{2k+1}
 \end{aligned}$$

Notice that  $f^{(2k+1)}(0) = (-1)^k$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}.$$

Let  $a_k = \frac{(-1)^k x^{2k+1}}{(2k+1)!}$

$$\begin{aligned}
 \left| \frac{a_{k+1}}{a_k} \right| &= \left| \frac{(-1)^{k+1} x^{2k+3}}{(2k+3)!} \cdot \frac{(2k+1)!}{(-1)^k x^{2k+1}} \right| \\
 &= x^2 \frac{1}{(2k+2)(2k+3)} \xrightarrow{k \rightarrow \infty} 0
 \end{aligned}$$

$\therefore R = \infty.$

Example: Find the Maclaurin series for  $\cos x$ .

$$\begin{aligned}
 \cos x &= \frac{d}{dx} (\sin x) = \frac{d}{dx} \left( \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \right) \\
 &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}
 \end{aligned}$$

Example: Represent  $\sin x$  as the sum of its Taylor series centered at  $x = \frac{\pi}{3}$ .

Sln:

$$\begin{aligned} f(x) &= \sin x & f\left(\frac{\pi}{3}\right) &= \frac{\sqrt{3}}{2} \\ f'(x) &= \cos x & f'\left(\frac{\pi}{3}\right) &= \frac{1}{2} \\ f''(x) &= -\sin x & f''\left(\frac{\pi}{3}\right) &= -\frac{\sqrt{3}}{2} \\ f'''(x) &= -\cos x & f'''\left(\frac{\pi}{3}\right) &= -\frac{1}{2} \end{aligned}$$

$$f\left(\frac{\pi}{3}\right) + \frac{f'\left(\frac{\pi}{3}\right)}{1!} \left(x - \frac{\pi}{3}\right) + \frac{f''\left(\frac{\pi}{3}\right)}{2!} \left(x - \frac{\pi}{3}\right)^2 + \dots$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2 \cdot 1!} \left(x - \frac{\pi}{3}\right) - \frac{\sqrt{3}}{2 \cdot 2!} \left(x - \frac{\pi}{3}\right)^2 - \frac{1}{2 \cdot 3!} \left(x - \frac{\pi}{3}\right)^3 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2 (2n+1)!} \left(x - \frac{\pi}{3}\right)^{2n+1} + \sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{3}}{2 (2n)!} \left(x - \frac{\pi}{3}\right)^{2n}$$

$\underbrace{\frac{f^{(2n+1)}\left(\frac{\pi}{3}\right)}{2 (2n+1)!} = (-1)^n \cdot \frac{1}{2}}_{\text{odd terms}}$        $\underbrace{\frac{f^{(2n)}\left(\frac{\pi}{3}\right)}{2 (2n)!} = (-1)^n \frac{\sqrt{3}}{2}}_{\text{even terms}}$